

1 Highlights

2 **Interpolation versus generalization in machine learning prediction** 3 **of micropollutant adsorption on layered double hydroxides**

4 [AUTHORS TO BE ADDED]

- 5 • 1,342 LDH adsorption records from 59 studies evaluated leakage-aware
- 6 • Random splits overstate ML skill: R^2 0.85 within studies, near zero
7 across
- 8 • Leave-one-study-out audit quantifies transfer for every source study
- 9 • Conformal intervals certify interpolation; new-study intervals widen 3–
10 $4\times$
- 11 • $\text{pH}-\text{pH}_{\text{pzc}}$ feature links model behavior to electrostatics

12 Interpolation versus generalization in machine learning
13 prediction of micropollutant adsorption on layered
14 double hydroxides

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16 **Abstract**

17 Micropollutants such as pharmaceuticals, pesticides, and personal-care prod-
18 ucts are detected in water bodies worldwide, and layered double hydroxides
19 (LDHs) are widely studied as tunable adsorbents for their removal. Machine
20 learning (ML) models trained on literature-compiled datasets increasingly
21 report near-perfect prediction of adsorption performance, yet the evaluation
22 protocols behind these reports rarely account for the structure of the under-
23 lying data, in which many experimental records originate from few source
24 studies. We compiled 1,342 adsorption experiments for LDH–micropollutant
25 systems from 59 published studies and evaluated seven regression models (lin-
26 ear regression, k -nearest neighbors, neural network, Cubist, random forest,
27 XGBoost, and histogram gradient boosting) under a leakage-aware proto-
28 col that separates two questions: interpolation within studied systems and
29 generalization to unseen studies. With duplicate records removed and all
30 preprocessing fitted on training folds only, the best model predicted removal
31 ratio (RR%) for studied systems at new operating conditions with R^2 of 0.87
32 and RMSE of 10 percentage points. Under leave-one-study-out evaluation,
33 however, the median per-study error rose to 14–16 percentage points, two

34 to three times the interpolation error, and only 20 of the 59 studies were
35 predicted with accuracy resembling the interpolation regime, showing that
36 random-split protocols overstate the screening ability of literature-trained
37 models by a wide margin; the models still reduce the error of descriptor-free
38 baselines by roughly 40 percent, so some signal does transfer. Split-conformal
39 prediction intervals certified the interpolation task (93% empirical coverage
40 at a 90% nominal level, with ± 19 -point intervals), but for unseen studies the
41 intervals had to widen to ± 68 points to maintain coverage, and a common
42 distance-to-training applicability score carried no information about realized
43 error in either task. Chemistry-informed features, including the electrostatic
44 driving force $\text{pH} - \text{pH}_{\text{pzc}}$, changed the errors little yet yielded feature attribu-
45 tions consistent with electrostatic and hydrophobic adsorption mechanisms.
46 The results delimit what literature-trained adsorption models can currently
47 support in practice: condition optimization for studied systems, but not yet
48 the screening of new adsorbent–pollutant combinations.

49 *Keywords:* layered double hydroxide, micropollutant, adsorption, machine
50 learning, data leakage, uncertainty quantification

51 Nomenclature

LDH	layered double hydroxide
RR%	removal ratio (percent)
C_i	initial concentration (mg/L)
AD	adsorbent dose (g/L)
pH _{pzc}	pH at the point of zero charge
BET	Brunauer–Emmett–Teller surface area (m ² /g)
PV	pore volume (cm ³ /g)
APD	average pore diameter (nm)
d_{003}	basal interlayer spacing (nm)
52 log K_{ow}	octanol–water partition coefficient
ML	machine learning
LR, KNN, ANN	linear regression, k -nearest neighbors, neural network
RF, XGB, HistGB	random forest, XGBoost, histogram gradient boosting
LOSO	leave-one-study-out
(T1), (T2)	interpolation task, cross-study generalization task
MAE, RMSE	mean absolute error, root mean square error
SHAP	Shapley additive explanations
CV	cross-validation

53 1. Introduction

54 Micropollutants, including pharmaceuticals, pesticides, endocrine disruptors,
55 and personal-care products, are detected at ng/L to µg/L levels in
56 surface water, groundwater, and treated effluent worldwide (TODO, 0000d).
57 Conventional treatment trains remove many of these compounds only par-

58 tially, and their chronic ecological and human-health effects have placed
 59 micropollutant control on regulatory agendas in the European Union, the
 60 United States, and Asia (TODO, 0000e). Adsorption is among the most
 61 widely deployed polishing steps because it requires no reagent input, pro-
 62 duces no transformation by-products, and operates across wide concentration
 63 ranges (TODO, 0000b).

64 Layered double hydroxides (LDHs) are a class of anionic clays with the
 65 general formula $[M_{1-x}^{2+}M_x^{3+}(\text{OH})_2]^{x+}[A^{n-}]_{x/n} \cdot m\text{H}_2\text{O}$, whose metal composi-
 66 tion, interlayer anion, and calcination state can be tuned to target specific
 67 pollutants (TODO, 0000c). Hundreds of LDH variants have been synthesized
 68 and tested against individual micropollutants, and each such study reports
 69 removal under its own grid of pH, temperature, contact time, adsorbent dose,
 70 and initial concentration. The resulting literature is large but fragmented: no
 71 single laboratory can cover the combinatorial space of metal pairs, synthesis
 72 routes, and pollutants experimentally.

73 Isotherm and kinetic models describe each of these systems in isolation
 74 and offer no mechanism for pooling evidence across studies. Machine learning
 75 (ML) offers a natural response to this fragmentation. By compiling published
 76 experiments into a single dataset, ML models have been trained to predict
 77 adsorption capacity or removal ratio from material, pollutant, and operat-
 78 ing descriptors, with reported test R^2 values frequently above 0.9 (TODO,
 79 0000f,g,a). Such models promise two distinct services: interpolating the be-
 80 havior of an already-studied adsorbent–pollutant system at untested condi-
 81 tions, and screening new systems before synthesis. The second service is the
 82 one most frequently invoked to motivate these studies.

83 However, we argue that the evaluation protocols behind these reports do
84 not support the screening claim. Firstly, literature-compiled datasets are
85 strongly clustered: in our compilation, 1,342 experimental records originate
86 from only 59 source studies, and within one study the material and pollutant
87 descriptors are constant while only operating conditions vary. A random
88 record-level train–test split therefore places near-duplicate records on both
89 sides, and the test score measures within-study interpolation while being
90 reported as generalization. Secondly, preprocessing steps such as imputa-
91 tion, scaling, and feature selection are commonly fitted on the full dataset
92 before splitting, which leaks test-set information into training (Kapoor and
93 Narayanan, 2023). Thirdly, point predictions are reported without uncer-
94 tainty estimates or applicability bounds, leaving practitioners no way to know
95 when a prediction should be trusted.

96 To address the above challenges, we evaluate ML models for micropollu-
97 tant adsorption on LDHs under a protocol that closes each of these leakage
98 pathways. In essence, our idea is to treat the source study as the unit of
99 generalization: we evaluate every model twice, once under a duplicate-free
100 record-level split that answers the interpolation question, and once under
101 grouped, leave-one-study-out protocols that answer the screening question,
102 with all preprocessing confined to training folds in both cases. We comple-
103 ment point predictions with split-conformal intervals and a distance-based
104 applicability-domain score, and we examine whether chemistry-informed fea-
105 tures improve cross-study transfer.

106 More specifically, this study aims to (1) quantify the gap between inter-
107 polation and cross-study generalization for seven regression models on 1,342

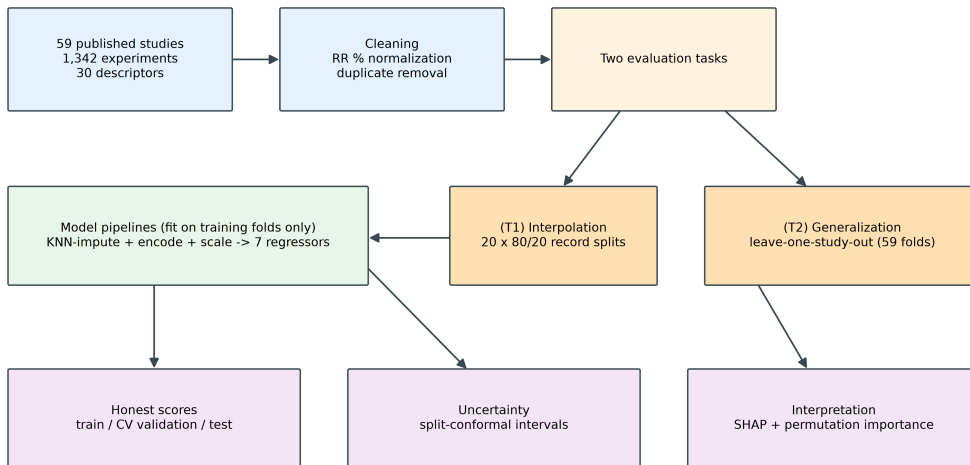


Figure 1: Evaluation workflow

LDH-micropollutant records from 59 studies; (2) equip the predictions with distribution-free uncertainty intervals and an applicability-domain flag, and test their reliability under both tasks; and (3) evaluate chemistry-informed features, including the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, and read the resulting feature attributions against known adsorption mechanisms. To the best of our knowledge, this is the first study of LDH adsorption modeling that separates these two prediction tasks explicitly and reports what literature-trained models can, and cannot yet, deliver for practice.

2. Materials and methods

Figure 1 summarizes the workflow: compilation and cleaning of the literature data, the two-task evaluation protocol, leakage-controlled model pipelines, and the uncertainty and interpretation layers.

120 2.1. Data curation

121 We compiled 1,342 batch-adsorption experiments for LDH–micropollutant
 122 systems from 59 peer-reviewed studies. Each record holds 30 descriptors in
 123 four groups: micropollutant properties (compound identity, $\log K_{ow}$, molecu-
 124 lar weight, the Abraham solvation parameters E , S , A , B , V , and topological
 125 polar surface area), LDH composition and synthesis (divalent and trivalent
 126 metal, molar ratio, synthesis route, hydrothermal temperature and time, cal-
 127 cination conditions, and composition-averaged elemental descriptors), mate-
 128 rial texture (interlayer spacing d_{003} , BET surface area, pore volume, average
 129 pore diameter, and point of zero charge pH_{pzc}), and operating conditions
 130 (pH, temperature, contact time, adsorbent dose, and initial concentration).
 131 Table 1 summarizes the numeric descriptors; the prediction target is the
 132 removal ratio (RR%).

133 **Cleaning.** Values stored as fractions were converted to percent by a
 134 range-aware rule that leaves already-percent data untouched, and category
 135 labels were normalized (e.g., merging spelling variants of *co-precipitation*).
 136 Records identical in every descriptor were reduced to a single occurrence,
 137 since duplicates spanning a train–test boundary would let the test be an-
 138 swered by lookup; deduplication removed 83 records. Missing texture val-
 139 ues were imputed by a k -nearest-neighbor imputer computed in standardized
 140 space and mapped back to physical units; the imputer, like every fitted trans-
 141 form in this work, was trained on training folds only. The source study of
 142 each record is retained as a grouping key and is never used as a model input.
 143 Details of the cleaning rules are in Text S1.

144 *2.2. Evaluation protocol*

145 **Two prediction tasks.** We evaluate every model under two protocols
146 that correspond to the two services expected from literature-trained models.
147 Task (T1), *within-system interpolation*, asks how well a model predicts RR%
148 for a studied adsorbent–pollutant system at new operating conditions. Task
149 (T2), *cross-study generalization*, asks how well the model predicts systems it
150 has never seen. Intuitively, (T1) allows the model to have seen the material
151 and the pollutant at other conditions, while (T2) does not.

152 **Repeated splits for (T1).** A single random split can be lucky or un-
153 lucky, so (T1) is measured over 20 seeded 80/20 record-level splits of the
154 duplicate-free data, and scores are reported as mean $\pm$ standard deviation
155 across repeats. Hyperparameters are tuned once (Section 2.3) and held fixed
156 across repeats, so the spread reflects sampling variability rather than tuning
157 noise.

158 **Leave-one-study-out audit for (T2).** Each of the 59 studies s is
159 held out once; the model f_{-s} is refit on the remaining 58 studies with fixed
160 hyperparameters and scored on the held-out records:

$$\text{MAE}_s = \frac{1}{n_s} \sum_{i \in s} |y_i - f_{-s}(\mathbf{x}_i)|. \quad (1)$$

161 Reporting the distribution of MAE_s across studies, rather than one pooled
162 score, exposes which systems transfer and which do not. A learning curve
163 over the number of training studies tests whether cross-study error is limited
164 by data volume or by data diversity.

165 **Naive baselines.** To make the error magnitudes interpretable, we score
166 three predictors that use no descriptors: the global training-mean; the per-
167 pollutant mean (the average RR% of the same compound in the training

168 data, falling back to the global mean for unseen compounds); and, for (T1)
169 only, the per-study mean. A model is useful for screening only if it beats
170 these lookups on unseen studies.

171 **Leakage control.** Imputation, scaling, and categorical encoding are
172 embedded in each model pipeline and fitted on training folds only. Hyperpa-
173 rameter tuning uses cross-validation folds that differ from the folds used for
174 reporting, and grouped tasks use group-aware folds throughout. These con-
175 trols address the leakage mechanisms documented for applied ML (Kapoor
176 and Narayanan, 2023), which we found to inflate reported skill substantially
177 when absent (Section 3.3).

178 *2.3. Regression models*

179 We compare seven regressors spanning the model families used in the
180 adsorption-ML literature, chosen so that their inductive biases differ.

181 **Linear regression (LR)** fits one additive coefficient per descriptor. It
182 serves as the interpretable floor: any nonlinearity or interaction in the ad-
183 sorption response appears as a gap between LR and the flexible models.

184 **k -nearest neighbors (KNN)** predicts the average RR% of the k most
185 similar training records in standardized feature space (Cover and Hart, 1967),
186 with k grid-searched between 1 and 9. KNN has no functional form at all;
187 its performance measures how far pure similarity lookup carries prediction,
188 which makes it a diagnostic for near-duplicate structure in the data.

189 **Artificial neural network (ANN)** is a single-hidden-layer perceptron
190 with 32 rectified linear units trained by Adam (Rumelhart et al., 1986),
191 matching the compact architectures used in prior adsorption studies. Larger
192 architectures were not explored because the dataset holds fewer than 10^3

193 effective degrees of freedom per feature.

194 **Cubist** builds a rule set whose leaves hold linear models and smooths
195 predictions with training-instance neighbors (Quinlan, 1992); committees
196 and neighbors are grid-searched. Unlike constant-leaf trees, its leaf-linear
197 structure can extrapolate moderately beyond the training range of a rule, a
198 property that becomes relevant under (T2).

199 **Random forest (RF)** averages deep decorrelated trees grown on boot-
200 strap samples (Breiman, 2001), providing a variance-reduced piecewise-constant
201 baseline among the ensembles.

202 **XGBoost** fits gradient-boosted trees with second-order updates and reg-
203 ularized leaf weights (Chen and Guestrin, 2016). **Histogram gradient**
204 **boosting (HistGB)** is the scikit-learn implementation of the same fam-
205 ily with histogram-binned split finding (Ke et al., 2017; Pedregosa et al.,
206 2011); both are the strongest current defaults for tabular data.

207 Tree-ensemble hyperparameters were tuned with the TPE sampler of Op-
208 tuna over 30 trials (Akiba et al., 2019) on cross-validation folds of the training
209 data. Scale-sensitive models (LR, KNN, ANN, Cubist) receive a standardizer
210 inside their pipeline; tree ensembles operate on raw feature scales. Search
211 spaces and selected values are in Table S1; all splits, folds, and samplers are
212 seeded for reproducibility.

213 2.4. Uncertainty and applicability domain

214 **Split-conformal intervals.** We wrap the best model in split-conformal
215 prediction (Vovk et al., 2005; Lei et al., 2018). A calibration subset of n
216 records is withheld from training, absolute residuals $r_i = |y_i - \hat{y}_i|$ are com-

217 puted on it, and the interval half-width is their conformal quantile

$$q = \text{Quantile}\left(\{r_i\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right), \quad (2)$$

218 so that each test prediction is reported as $[\hat{y}-q, \hat{y}+q]$ with nominal coverage
 219 $1-\alpha=0.9$. For task (T2) the calibration subset is drawn as whole studies,
 220 so the calibration residuals reflect the new-study setting. Conformal validity
 221 assumes exchangeability between calibration and test units, which grouped
 222 data satisfy only approximately; we therefore report empirical coverage for
 223 both tasks rather than assuming it.

224 **Applicability domain.** As a candidate trust flag we compute, for each
 225 test record, the mean standardized distance to its five nearest training records
 226 in the preprocessed feature space. Records far from the training distribution
 227 should receive predictions with caution; we validate the flag by correlating it
 228 with realized absolute error rather than assuming it works.

229 2.5. Feature engineering and attribution

230 Beyond the raw descriptors, we construct three features with mechanistic
 231 meaning: the electrostatic driving force $\text{pH}-\text{pH}_{\text{pzc}}$, whose sign determines
 232 whether the LDH surface is positively or negatively charged relative to the
 233 solution; the initial loading ratio C_i/dose ; and the interaction $\log K_{\text{ow}} \times$
 234 $(\text{pH}-\text{pH}_{\text{pzc}})$ coupling hydrophobicity with surface charge. Both tasks are
 235 re-evaluated with and without these features. Feature attributions are com-
 236 puted with SHAP (Lundberg and Lee, 2017) on the held-out test set using
 237 the fitted models, complemented by permutation importance ranks across all
 238 seven models and one-dimensional partial dependence profiles, and the re-
 239 sulting patterns are read against the expected electrostatic and hydrophobic

240 mechanisms.

241 2.6. Performance metrics

242 Model skill is reported as the coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (3)$$

243 the root mean square error, $\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2}$, and the mean abso-
244 lute error, $\text{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$, each computed on three bases with distinct
245 meanings: the training records (in-sample fit), cross-validation folds of the
246 training set (validation), and the held-out test records of the task at hand.
247 All three are reported to keep the in-sample ceiling and the generalization
248 estimate visibly separate. Wall-clock training and prediction times are re-
249 ported in Section 3.6.

250 3. Results and discussion

251 Section 3.1 first characterizes the compiled dataset. The modeling ex-
252 periments then answer four research questions. (RQ1) How accurately do
253 the models predict RR% for studied systems at new operating conditions?
254 (RQ2) Does this accuracy transfer to systems from unseen studies? (RQ3) Can
255 the uncertainty of individual predictions be quantified reliably under both
256 tasks? (RQ4) Which factors govern the predictions, and are they chemically
257 plausible?

258 3.1. Dataset characteristics

259 Figure 2 summarizes the compiled data. Removal ratios span the full
260 0–100% range with a median above 50%, but the distribution differs strongly

261 between pollutants and between LDH metal pairs (Fig. 2a,b), reflecting both
 262 genuine chemistry and the uneven attention different systems have received
 263 in the literature. The study-size distribution is heavily skewed: the largest
 264 single study contributes 220 of the 1,342 records, the median study only
 265 17, and a random record-level sample is therefore dominated by a hand-
 266 ful of intensively measured systems. Spearman correlations among the nu-
 267 meric descriptors (Fig. S1) show the expected blocks: the Abraham solvation
 268 parameters correlate strongly with molecular weight and with each other,
 269 and the composition-averaged elemental descriptors correlate with the metal
 270 identities they are derived from. This redundancy motivates the correlated-
 271 descriptor experiment in Section S3 and, more fundamentally, explains why
 272 record-level splits leak: within one study, most descriptor columns are con-
 273 stant, so any two records of the same study are near-duplicates in feature
 274 space. Missingness is confined to the synthesis and texture blocks (Table 1):
 275 calcination conditions are unreported for two thirds of the records, and pH_{pzc}
 276 and pore volume for roughly a quarter, which is why imputation quality mat-
 277 ters and why texture-based mechanistic claims from compiled data warrant
 278 caution.

279 3.2. Within-system interpolation (RQ1)

280 Table 2 answers (RQ1) by comparing the seven models and the naive
 281 baselines under task (T1), reported as mean $\pm$ SD over 20 seeded splits.
 282 The tree ensembles lead: XGBoost and HistGB reach test R^2 of 0.84–0.85
 283 with RMSE near 11 percentage points, followed by Cubist and RF, while LR
 284 trails far behind. This is because $\text{RR}\%$ responds nonlinearly to the operating
 285 conditions, with thresholds in pH and dose that additive linear terms cannot

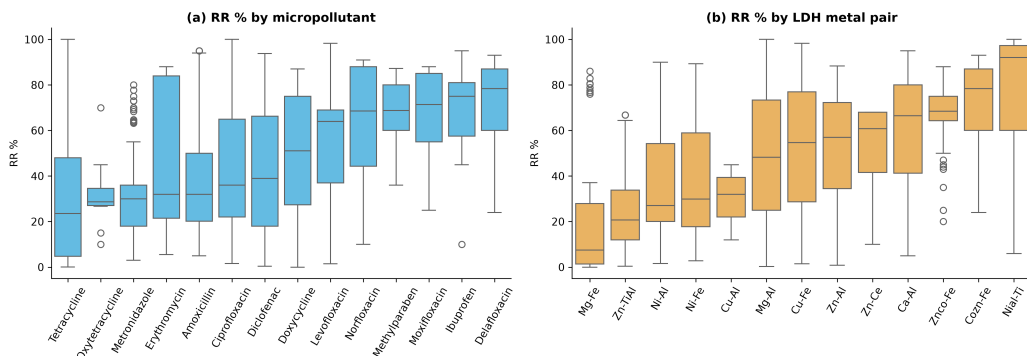


Figure 2: Removal-ratio distributions by pollutant and by metal pair

represent. The baselines put these numbers in perspective. The per-study mean, which uses no descriptors at all, already attains R^2 of 0.43 (MAE 16.7): nearly half of the variance a record-level split appears to explain is available from system identity alone, and LR barely improves on it. The ensembles' further gain to R^2 of 0.85 therefore reflects genuine modeling of the operating conditions within known systems. The in-sample R^2 of the ensembles approaches 0.99, and the gap to the test score is the expected overfitting margin of high-capacity learners rather than a protocol artifact; the cross-validation and test scores agree closely. Figure 3 shows the parity and residual plots of the best model on the reference split. It is worth noting that these scores are already 2–3 RMSE points more conservative than a protocol without duplicate removal on the same data, since deduplication alone eliminates records whose test values could be looked up verbatim from training.

300 *3.3. Cross-study generalization (RQ2)*

301 Figure 4 answers (RQ2) with the leave-one-study-out audit. The picture
302 changes sharply: the median per-study MAE of the best models rises to 14–
303 16 percentage points, two to three times their interpolation error, and the
304 per-study R^2 is positive for only 4 of the 32 studies large enough to evaluate
305 it. This is because the models exploit descriptor combinations that identify
306 a material–pollutant system rather than transferable structure–activity rela-
307 tions; when all records of a system are withheld, that shortcut disappears.
308 The baseline comparison adds an important nuance. Descriptor-free lookups
309 transfer even worse: the global training-mean reaches a median per-study
310 MAE of 24 points and the per-pollutant mean of 25, so the models do reduce
311 the naive error by roughly 40 percent, and Cubist beats the per-pollutant
312 lookup in 41 of the 59 studies. The models therefore track the level differences
313 between systems while failing to resolve the condition response within an un-
314 seen system, which is exactly why the per-study R^2 (measured against each
315 study’s own variance) stays negative even as the MAE gain over the baselines
316 is large. Screening applications get a coarse ranking signal from these mod-
317 els, not engineering precision. The per-study distribution in Fig. 4b is wide
318 rather than uniformly poor: for Cubist, 20 of the 59 studies are predicted
319 with errors comparable to the interpolation regime (MAE below 6 points),
320 while 9 studies exceed 30 points. Notably, the model ranking also reorders:
321 Cubist, mid-table under interpolation, attains the lowest cross-study median,
322 which could be explained by its rule-plus-linear structure extrapolating more
323 gracefully than deep tree ensembles.

324 **Which systems transfer.** Figure 6 breaks the per-study errors down

325 by chemistry, and the pattern follows literature density. Ciprofloxacin stud-
 326 ies, which dominate the compilation (32 of 59 studies), are predicted with
 327 a median MAE of 5 points, and Zn–Al systems (28 studies) with 5 points,
 328 because each held-out study of these families leaves many close relatives in
 329 training. Sparsely covered chemistries fare far worse: tetracycline, diclofenac,
 330 and levofloxacin studies show median errors above 20 points, and Ni–Fe sys-
 331 tems above 30. Error also grows with the held-out study’s own scope: per-
 332 study MAE correlates with both the number of records and the within-study
 333 spread of RR% ($r \approx 0.45$ for each), meaning studies that explore wide con-
 334 dition ranges are precisely the ones an outside model cannot resolve. The
 335 learning curve in Fig. 5b is flat between 5 and 40 training studies, suggesting
 336 that cross-study error is limited by the diversity of compiled systems rather
 337 than by record count. This supports our central argument that random-split
 338 scores, including those in Table 2, must not be read as screening skill: the
 339 same models, the same data, and the same preprocessing yield near-zero skill
 340 when the test set contains genuinely new systems.

341 3.4. *Uncertainty and applicability domain (RQ3)*

342 Figure 7 answers (RQ3). Under interpolation, the split-conformal 90%
 343 intervals of Eq. (2) achieve empirical coverage of 93% with a half-width of
 344 19 percentage points, so predictions for studied systems can be certified at
 345 a practically useful precision. Under the new-study task, coverage is main-
 346 tained (99%) only because the study-grouped calibration inflates the half-
 347 width to 68 points, an interval spanning most of the 0–100 RR% range and
 348 therefore carrying little decision value. This is because residual magnitudes
 349 differ systematically between systems, so a single global quantile must cover

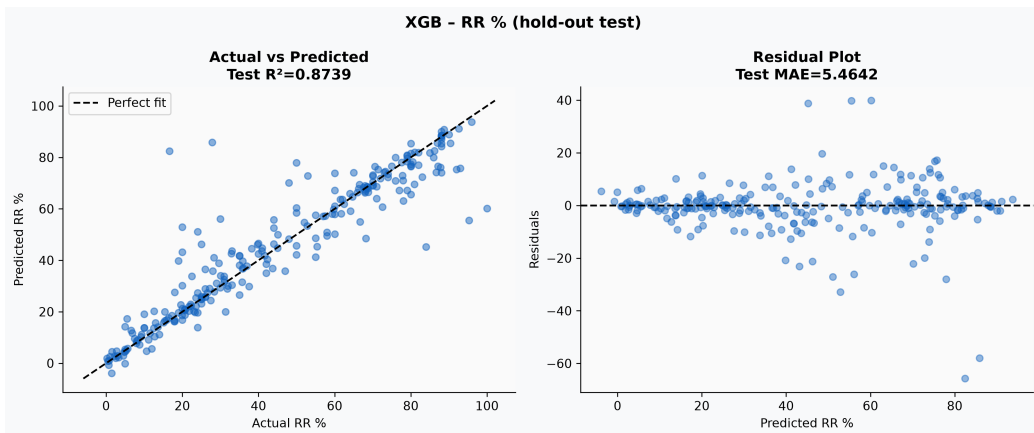


Figure 3: Parity and residual plots of the best model under interpolation

the worst of them. The distance-based applicability-domain flag does not compensate: the mean distance to the five nearest training records carries essentially no information about realized error in either task ($r \leq 0.06$, Fig. 7b), although such distance heuristics are widely used as trust flags. A key finding is therefore that the uncertainty tools inherit the transfer problem itself: interpolation predictions can be certified, whereas no construction we tested produces informative guarantees for unseen systems, and coverage claims must be validated empirically for the deployment task.

3.5. Feature effects and mechanistic reading (RQ4)

Figure 8 and Table 3 answer (RQ4). Adding the chemistry-informed features leaves both tasks essentially unchanged (interpolation MAE 5.6 versus 5.8 points; cross-study median 16.4 versus 16.0), indicating that the models already reconstruct these combinations internally from pH, pH_{pzc}, and dose. Their value here is interpretive rather than predictive.

Operating conditions dominate. Permutation-importance ranks on

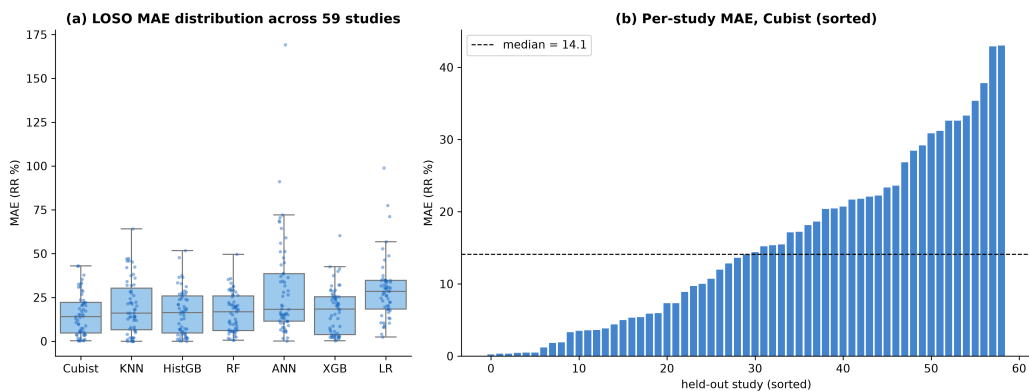


Figure 4: Leave-one-study-out audit across 59 studies

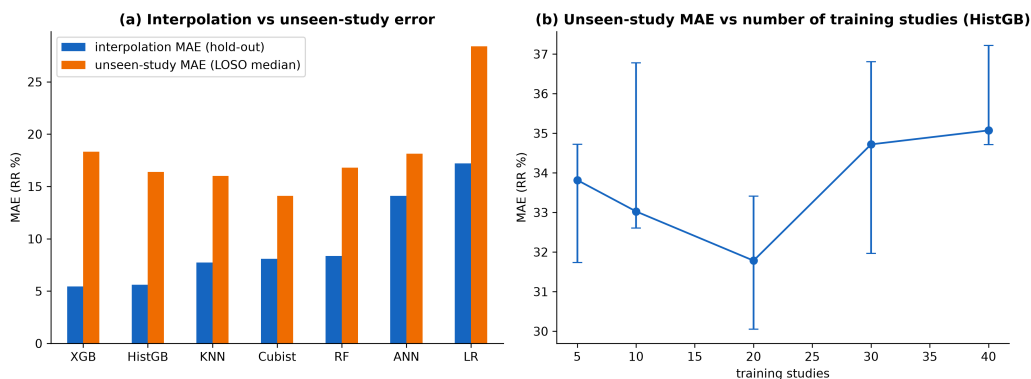


Figure 5: Interpolation versus unseen-study error and learning curve

365 the hold-out set (Table 3) agree across six of the seven models on the leading
 366 factors: adsorbent dose ranks first, followed by contact time, initial concen-
 367 tration, and pH, with the first material property (BET) at median rank six
 368 and the first pollutant descriptor ($\log K_{ow}$) at rank nine; only the additive
 369 LR deviates. The partial-dependence profiles of the three leading features
 370 (Fig. S2) show textbook adsorption behavior: removal rises steeply with
 371 dose up to roughly 2–5 g/L and then saturates, rises within the first hours
 372 of contact and plateaus at equilibrium, and declines with initial concentra-

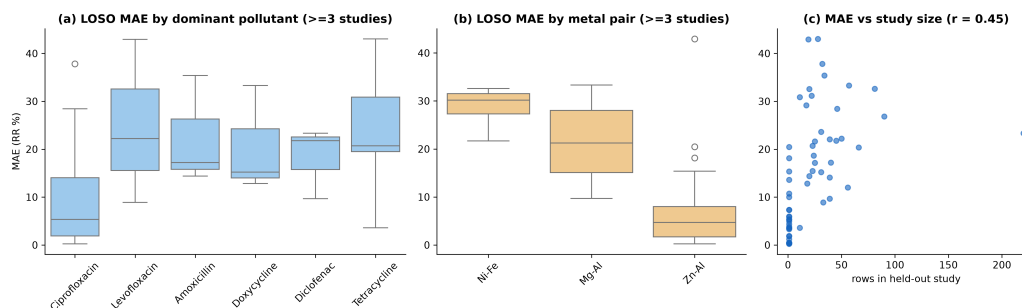


Figure 6: Cross-study error by pollutant, metal pair, and study size

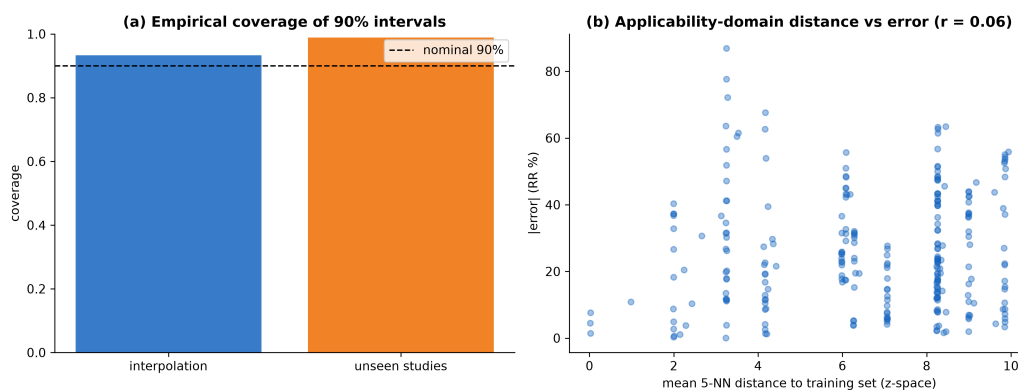


Figure 7: Conformal coverage and applicability-domain flag

tion beyond roughly 200 mg/L as fixed capacity is exhausted. Within known systems, the models therefore learn genuine condition–response physics. The same ranking, however, explains the transfer failure of Section 3.3 from a second direction: material and pollutant descriptors, the only information available about an unseen system, contribute comparatively little to the fitted models.

Mechanistic reading. The SHAP dependence of $\text{pH} - \text{pH}_{\text{pzc}}$ (Fig. 8a) is not smoothly monotone but threshold-like: contributions stay within a few points while the surface remains positively charged, and a sharp penalty

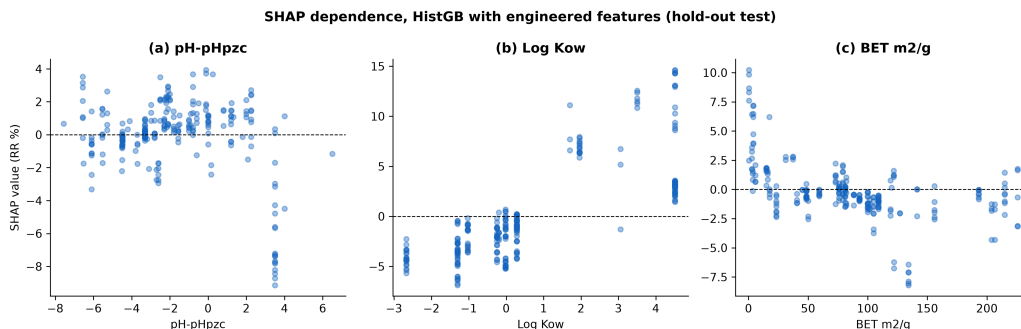


Figure 8: SHAP dependence of mechanism-linked features

of up to nine points appears once solution pH exceeds pH_{pzc} by more than
 about three units. Coloring this dependence by $\log K_{\text{ow}}$ (Fig. S3) shows the
 penalty concentrated among hydrophilic compounds, consistent with elec-
 trostatic repulsion of anionic species from a negatively charged surface while
 hydrophobic partitioning buffers the loss for lipophilic ones. $\log K_{\text{ow}}$ itself
 is the strongest chemical attribution (Fig. 8b), monotone from roughly -5
 points for the most hydrophilic compounds to $+15$ for the most lipophilic.
 BET surface area (Fig. 8c) is a cautionary counterexample: the model assigns
 positive contributions to records with low measured BET even though the
 raw association runs the other way (median $\text{RR}\%$ of 44 below $20 \text{ m}^2/\text{g}$ versus
 50 above), so this conditional attribution reflects the correlation structure of
 the compilation rather than a surface-area mechanism, and we do not inter-
 pret it mechanistically. Attributions for the composition-averaged elemental
 descriptors are small, consistent with the finding that current composition
 descriptors carry little transferable information between studies. [MECHA-
 NISM PARAGRAPH FLAGGED FOR DOMAIN REVIEW BY THE AU-
 THORS.]

399 3.6. Computation time

400 Training cost is not a differentiator at this data scale. On the reference
401 split, a single fit takes below one second for every model except XGBoost (7–
402 10 s across repeated measurements with the tuned 495-tree configuration),
403 and prediction takes below 0.1 s throughout (Table S3). The full leave-one-
404 study-out audit, 59 refits of all seven models, completes within minutes on a
405 workstation, so the honest grouped protocol carries no meaningful computa-
406 tional premium over the random-split protocol it replaces. Hyperparameter
407 search with Optuna dominates the overall budget and is performed once, not
408 per evaluation task.

409 3.7. Environmental implications and outlook

410 For practice, the results delimit two use cases sharply. Water-treatment
411 engineers optimizing operating conditions for an LDH–pollutant system that
412 has been measured before can rely on literature-trained models with a mean
413 absolute error near 6 percentage points of RR%, tightened further by confor-
414 mal intervals. Materials researchers seeking to screen new LDH formulations
415 or new pollutants face a harder trade: the models do carry transferable sig-
416 nal, reducing naive baseline errors by roughly 40 percent, but per-study skill
417 remains near zero, so predictions for unmeasured systems support coarse
418 prioritization at best and should be treated as hypotheses for experiment
419 rather than estimates. Reported R^2 values above 0.9 in prior compilations
420 (TODO, 0000f,a) are consistent with our record-level numbers and, by the
421 mechanism quantified here, likely overstate screening ability in the same way;
422 re-evaluation of published models under grouped protocols would clarify how
423 general this pattern is.

424 We see three concerns that temper the conclusions. First, the compila-
425 tion covers 59 studies, so the cross-study estimates themselves carry sampling
426 uncertainty; the per-study distribution, not a single mean, is the honest sum-
427 mary. The second concern is reporting heterogeneity between laboratories
428 (differing protocols for pH_{pzc} or BET), which acts as irreducible noise under
429 (T2) and may partly explain the transfer gap. The third concern is that our
430 descriptors, like those of prior compilations, describe the bulk material; in-
431 terlayer anion identity and surface morphology are rarely reported and may
432 carry the transferable signal that current models lack. In future work, richer
433 composition-structure descriptors, study-level random effects, and few-shot
434 calibration on a handful of measurements of a new system are the most
435 promising routes toward genuine screening capability.

436 4. Conclusions

437 We evaluated seven machine learning models for micropollutant adsorp-
438 tion on layered double hydroxides under a leakage-aware protocol that sepa-
439 rates within-system interpolation from cross-study generalization, using 1,342
440 experiments compiled from 59 studies. The models predict removal ratio for
441 studied systems at new operating conditions with test R^2 of 0.85 ± 0.03
442 and RMSE near 11 percentage points across 20 repeated splits, but the same
443 models show near-zero per-study skill when entire studies are held out, and a
444 leave-one-study-out audit locates the failure in a wide per-study error distri-
445 bution rather than a uniform degradation. Against descriptor-free baselines
446 the picture is more nuanced: the models reduce the naive cross-study error
447 by roughly 40 percent, so they do carry transferable signal, yet not enough

448 to certify predictions for an unmeasured system. Split-conformal intervals
449 certify interpolation predictions at ± 19 percentage points, whereas for un-
450 seen studies neither conformal calibration nor a distance-based applicability
451 score yields informative guarantees. Chemistry-informed features such as
452 $\text{pH} - \text{pH}_{\text{pzc}}$ leave the accuracy essentially unchanged but anchor the learned
453 behavior in electrostatic and hydrophobic mechanisms. In future work, we
454 plan to extend the compilation with interlayer-anion and morphology de-
455 scriptors, model study-level effects explicitly, and test few-shot calibration
456 so that a small number of measurements of a new system can recover the
457 accuracy currently reserved for studied ones.

458 **CRedit authorship contribution statement**

459 [TO BE COMPLETED BY THE AUTHORS.]

460 **Declaration of competing interest**

461 The authors declare that they have no known competing financial inter-
462 ests or personal relationships that could have appeared to influence the work
463 reported in this paper.

464 **Data availability**

465 The curated dataset, the full modeling pipeline (`ldh_pipeline.py`), and
466 the notebooks reproducing every table and figure are available at [REPOSI-
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Table 1: Summary statistics of the numeric descriptors

Descriptor	Median	Range	Missing (%)
<i>Micropollutant</i>			
$\log K_{ow}$	−0.02	−2.67–4.51	0
MW (g/mol)	331	152–734	0
$E / S / A / B / V$	1.81 / 2.10 / 0.70 / 1.70 / 2.30	–	0
TPSA (Å ²)	76.2	37.3–194.0	0
<i>Composition and synthesis</i>			
M(II)/M(III) ratio	3.0	0.5–8.0	0
Avg. atomic weight (g/mol)	48.1	24.8–73.7	0
Avg. d-electrons	4.25	0–8.89	0
Avg. Pauling electronegativity	1.59	1.30–1.89	0
Hydrothermal temp. (°C)	70	25–160	13.3
Hydrothermal time (h)	24	12–40	13.3
Calcination temp. (°C)	400	200–550	65.9
Calcination time (h)	4	1–12	65.9
<i>Material texture</i>			
d_{003} (nm)	0.77	0.25–1.13	2.5
BET (m ² /g)	82.0	0.4–225.7	7.9
PV (cm ³ /g)	0.10	0–1.46	25.3
APD (nm)	8.4	0.2–64.2	13.9
pH _{pzc}	7.5	5.5–11.6	24.6
<i>Operating conditions</i>			
pH	6.7	2.0–12.0	0
Temperature (°C)	25	10–55	0
Contact time (h)	1.5	0.02–192	0
Dose (g/L)	0.9	0.01–50	0
C_i (mg/L)	40	1–1688	0

Table 2: Model and baseline performance under within-system interpolation (T1), mean $\pm$ SD over 20 seeded 80/20 splits [VALUES FROM repeated_interp_summary.csv]

Model	Test R^2	Test MAE	Test RMSE
XGBoost	0.84 ± 0.03	6.3 ± 0.6	11.1 ± 1.0
HistGB	0.85 ± 0.03	6.3 ± 0.5	10.8 ± 1.0
KNN	0.73 ± 0.06	7.8 ± 0.8	14.4 ± 1.5
Cubist	0.79 ± 0.03	8.0 ± 0.6	12.7 ± 1.0
RF	0.78 ± 0.03	8.6 ± 0.5	13.0 ± 0.9
ANN	0.54 ± 0.05	14.4 ± 0.7	19.1 ± 0.9
LR	0.40 ± 0.04	17.4 ± 0.5	21.7 ± 0.7
Per-study mean	0.43 ± 0.05	16.7 ± 0.8	21.2 ± 0.9
Per-pollutant mean	0.18 ± 0.03	21.4 ± 0.6	25.4 ± 0.6
Global mean	-0.01 ± 0.01	24.9 ± 0.7	28.1 ± 0.6

Table 3: Permutation-importance ranks of the leading descriptors across the seven models (hold-out test set) [VALUES FROM perm_ranks_all_models.csv]

Descriptor	LR	KNN	ANN	Cubist	RF	XGB	HistGB
AD (dose)	27	1	29	1	1	1	1
Time	19	3	1	2	2	2	2
C_i	22	4	14	3	3	4	3
pH	25	2	4	5	7	5	4
BET	24	5	2	7	6	6	6
pH _{pzc}	26	7	19	9	4	7	5
d_{003}	8	6	8	8	12	8	10
Avg. electronegativity	15	13	7	15	9	9	8